

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how FitBuddy can scale to more users and data, and the team's roadmap for improving it.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single instance with an SQLite file database and local run	Cannot serve many simultaneous users; data lives on one machine	High
2	Depends on Google Gemini availability, quota and speed	Slow or failed plan generation when the service is busy or the quota is used	High
3	No user accounts; admin routes do not yet enforce the password	Weak privacy: user data could be viewed or deleted without login	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One Uvicorn process on a local machine	Several workers behind Nginx, deployed from the Docker image to a cloud service; asynchronous or background AI calls
2	Data Storage	SQLite file	Managed PostgreSQL with backups (change DATABASE_URL); indexes on user_id
3	Performance	Each request calls Gemini directly, no caching	Caching of common tips, rate limiting, a lighter model where suitable, request queue
4	Security	Secrets in .env; admin Basic auth not yet enforced on routes	Enforce admin authentication, add user accounts (JWT/OAuth), HTTPS, secrets manager

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Enforce admin login on all admin routes; user accounts and login; history of plan versions	Month 1	Protects user data and personalises the experience
Phase 3	Progress tracking, structured (JSON) plans, calendar reminders	Months 2 to 3	Higher engagement and easier plan display
Phase 4	Cloud deployment with PostgreSQL and CI/CD; multi-language support; wearable integration	Months 4 to 6	Supports more users and wider reach